

Censored data analysis

1 Introduction: censored data

In certain practical applications, we only have access to censored data. A censored variable has a significant fraction of the observations truncated above or below at some value. Examples of censored data include:

- Survival of patients (some patients may leave a medical programme before concluding it);
- Abandonment time (patience) of customers waiting in queue at some call center. It is censored by the time the person is served;
- Number of extramarital affairs (data collected, for example, by some magazine surveys);
- Expenditures on vacations;
- Education testing: if an exam happens to be too easy, a lot of people will get full marks, while if it is too hard, a lot of people may get zero marks.

We focus on two particular applications in social sciences and healthcare.

2 Censored data analysis in Social Sciences

Tool: Tobit model.

We analyze a dataset where one wants to model the time spent on extra-marital affairs using certain predictor variables like number of affairs, religiosity, age, education, etc. The data set contains many observations where the time spent in extra-marital affairs is zero, essentially acting as a censor for the response variable. A linear regression model would not be able to capture this relationship properly, hence we resort to using a specific statistical model called the Tobit model.

2.1 Dealing with censored data

2.1.1 Censored-dependent variable

Let's assume that a censored-dependent variable Y^* follows a normal distribution with mean μ and standard deviation σ , that is, $Y^* \sim \mathcal{N}(\mu, \sigma)$. Consider a censored value at C (say, capacity), then, the variable we observe is:

$$Y = \begin{cases} Y^*, & \text{if } Y^* \leq C, \\ C, & \text{otherwise.} \end{cases}$$

In this example, illustrated in Figure 1.1, the censored random variable is right-censored and the new variable is a mixture of continuous and discrete points.

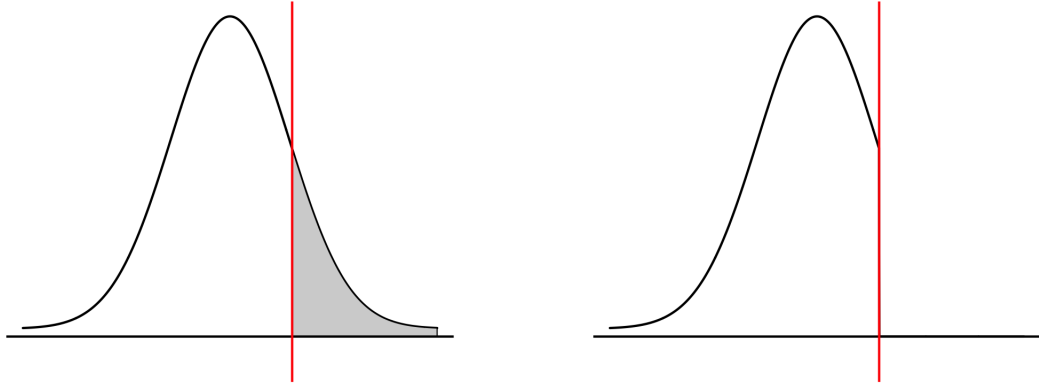


Figure 1.1: Illustration of a right-censored random variable(hypothetical). Left plot: distribution/density of the original variable (say demand for soccer tickets/sets); Right plot: observed demand based on ticket sales which is capped at the capacity of the stadium.

2.1.2 Censored regression (Tobit model)

James Tobin, in 1958, proposed a model that deals with censored regression problems. The model is referred to as the Tobit model (from Tobin and probit.) The model is based on the following regression:

$$\underbrace{Y_i^*}_{\substack{\text{Latent variable} \\ \text{(unobservable)}}} = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} + \epsilon_i, \quad \forall i = 1, \dots, n,$$

where $\{\beta_1, \dots, \beta_p\}$ are the model coefficients, $\{x_{i1}, \dots, x_{ip}\}$ the predictors for the i -th individual, n the number of observations, and ϵ_i the model error. The Tobit model assumes that $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$. The observable variable Y_i is left-censored at 0:

$$\underbrace{Y_i}_{\substack{\text{observable} \\ \text{variable}}} = \begin{cases} Y_i^*, & \text{if } Y_i^* \geq 0, \\ 0, & \text{if } Y_i^* < 0. \end{cases}$$

Since Y^* is normally distributed, Y has a continuous distribution over strictly positive values. In particular, the density of Y given $\mathbf{x}$ (the vector of predictors $x_1, \dots, x_p$) is the same as the density of Y^* given $\mathbf{x}$ for positive values. Further,

$$\begin{aligned} \mathbb{P}(Y = 0 | \mathbf{x}) &= \mathbb{P}(Y^* < 0 | \mathbf{x}) = \mathbb{P}(\epsilon < -\mathbf{x}^\top \boldsymbol{\beta} | \mathbf{x}) \\ &= \mathbb{P}(\epsilon/\sigma < -\mathbf{x}^\top \boldsymbol{\beta}/\sigma | \mathbf{x}) \\ &= \Phi(-\mathbf{x}^\top \boldsymbol{\beta}/\sigma) = 1 - \Phi\left(\frac{\mathbf{x}^\top \boldsymbol{\beta}}{\sigma}\right), \end{aligned}$$

where we absorbed the intercept β_0 into a vector of coefficients $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_p)$, $\mathbf{x} = (1, x_1, \dots, x_p)$ and $\Phi(\cdot)$ is the cumulative distribution function of a normal variable. This expression holds because ϵ/σ has a standard normal distribution and is independent of $\mathbf{x}$. Therefore, if $(\mathbf{x}_i, y_i)$ is a random draw from the population, the density of Y given $\mathbf{x}_i$ is:

$$f_{Y|\mathbf{X}}(y_i|\mathbf{x}_i) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2}{2\sigma^2}} = \frac{1}{\sigma} \phi\left(\frac{y_i - \mathbf{x}_i^\top \boldsymbol{\beta}}{\sigma}\right), \quad y_i > 0, \quad \text{and,}$$

$$\mathbb{P}(Y = 0|\mathbf{x}_i) = \Phi\left(-\frac{\mathbf{x}_i^\top \boldsymbol{\beta}}{\sigma}\right) = 1 - \Phi\left(\frac{\mathbf{x}_i^\top \boldsymbol{\beta}}{\sigma}\right).$$

where ϕ is the standard normal density function. From the previous two equations, we can obtain the log-likelihood function for each observation i :

$$l_i(\boldsymbol{\beta}, \sigma) = \mathbf{1}(Y_i = 0) \log \left[1 - \Phi\left(\frac{\mathbf{x}_i^\top \boldsymbol{\beta}}{\sigma}\right) \right] + \mathbf{1}(Y_i > 0) \log \left[\frac{1}{\sigma} \phi\left(\frac{y_i - \mathbf{x}_i^\top \boldsymbol{\beta}}{\sigma}\right) \right].$$

Notice how this depends on σ , the standard deviation of ϵ , as well as on $\boldsymbol{\beta}$. The log-likelihood for a random sample of size n is obtained by summing $l_i(\cdot)$ across all i . The maximum likelihood estimates of $\boldsymbol{\beta}$ and σ are obtained by maximizing the log-likelihood. This generally requires numerical methods.

2.1.3 Interpreting the model output

From equation $Y^* = \mathbf{x}^\top \boldsymbol{\beta} + \epsilon$, we see that the parameter β_j measure the partial effects of x_j on $\mathbb{E}(Y^*|\mathbf{x})$, where Y^* is the latent variable. The variable we want to explain is Y , as this is the observed outcome (such as hours worked or amount of charitable contributions).

In Tobit models, two expectations are of particular interest: $\mathbb{E}(Y|Y > 0, \mathbf{x})$, which is sometimes called the *conditional expectation* because it is conditional on $Y > 0$, and $\mathbb{E}(Y|\mathbf{x})$, which is, unfortunately, called the *unconditional expectation* (both expectations are conditional on the explanatory variables). The expectation $\mathbb{E}(Y|Y > 0, \mathbf{x})$ tells us, for given values of $\mathbf{x}$, the expected value of Y for the subpopulation where Y is positive. Given $\mathbb{E}(Y|Y > 0, \mathbf{x})$, we can easily find $\mathbb{E}(Y|\mathbf{x})$:

$$\mathbb{E}(Y|\mathbf{x}) = \mathbb{P}(Y > 0|\mathbf{x}) \cdot \mathbb{E}(Y|Y > 0, \mathbf{x}) = \Phi\left(\frac{\mathbf{x}^\top \boldsymbol{\beta}}{\sigma}\right) \cdot \mathbb{E}(Y|Y > 0, \mathbf{x}).$$

Skipping some derivations, $\mathbb{E}(Y|Y > 0, \mathbf{x})$ can be expressed as $\mathbb{E}(Y|Y > 0, \mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta} + \sigma \lambda\left(\frac{\mathbf{x}^\top \boldsymbol{\beta}}{\sigma}\right)$, which shows that the expected value of Y conditional on $Y > 0$ is equal to $\mathbf{x}^\top \boldsymbol{\beta}$ plus a strictly positive term, which is σ times $\lambda\left(\frac{\mathbf{x}^\top \boldsymbol{\beta}}{\sigma}\right)$.

Combining the last two expressions, we get:

$$\mathbb{E}(Y|\mathbf{x}) = \Phi\left(\frac{\mathbf{x}^\top \boldsymbol{\beta}}{\sigma}\right) \cdot \left[\mathbf{x}^\top \boldsymbol{\beta} + \sigma \lambda\left(\frac{\mathbf{x}^\top \boldsymbol{\beta}}{\sigma}\right) \right] = \Phi\left(\frac{\mathbf{x}^\top \boldsymbol{\beta}}{\sigma}\right) \mathbf{x}^\top \boldsymbol{\beta} + \sigma \phi\left(\frac{\mathbf{x}^\top \boldsymbol{\beta}}{\sigma}\right),$$

where the second equality follows because $\Phi\left(\frac{\mathbf{x}^\top \boldsymbol{\beta}}{\sigma}\right) \lambda\left(\frac{\mathbf{x}^\top \boldsymbol{\beta}}{\sigma}\right) = \phi\left(\frac{\mathbf{x}^\top \boldsymbol{\beta}}{\sigma}\right)$. This equation shows that when Y follows a Tobit model, $\mathbb{E}(Y|\mathbf{x})$ is a nonlinear function of $\mathbf{x}$ and $\boldsymbol{\beta}$.

3 Survival analysis in healthcare

Tool: Kaplan-Meier estimator, Cox proportional hazard model.

Survival data are common in different domains, such as medical research or transportation engineering. They generally describe situations in which an event of interest (e.g., the death of a patient or the failure of a mechanical component) can happen at different times. Descriptive analytics can help estimate the survival rate of patients or predict the timing of failures.

3.1 Overview

Duration (or survival) analysis aims at analyzing the expected duration of time until a specific *event* happens. In other words, the *duration* is the time from the beginning of an observation period to the event of interest (or the end of the study or the loss of contact/withdrawal from the study). Consider the example illustrated in Figure 1.2: at time t , we observe the status of a patient after a heart transplant. At that time, the patient might be dead or alive. If the patient is alive (so he/she does not have an event during the observation time), we are dealing with a *censored* observation (the subject is censored in the sense that nothing is observed or known about that subject after the time of censoring). So, the survival time for the alive patient is at least t , but not exactly t . Our interest is to describe the *survival function* $S(t)$, which expresses the probability that a subject survives longer than time t .

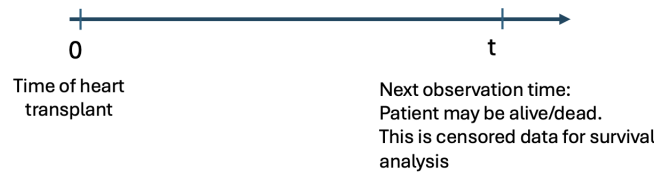


Figure 1.2: Survival data for patients after a surgery.

Another example of duration analysis is a government studying the length of unemployment for a group of citizens. In such case, the event of interest could be the time at which a citizen finds a job. Duration analysis finds application in several sectors, such as transport engineering, where analysts are typically interested in estimating the length of time until failure of a certain mechanical component. Note that duration times are by definition nonnegative.

3.2 Dealing with survival data

3.2.1 Survival function and hazard rate

Let's denote with T a continuous random variable that represents the waiting time until the occurrence of an event (e.g., death). T has a probability density function $f(t)$ and cumulative density function $F(t) = \mathbb{P}(T \leq t)$ (giving the probability that the event has occurred by duration t). The *survival function* $S(t)$ can thus be defined as

$$S(t) = \mathbb{P}(T > t) = 1 - F(t).$$

The survival function represents the probability of being alive just before duration t , that is, the probability that the event of interest has not occurred by duration t .

One important question in analysis of duration data is: given that an event has lasted until t , what is the probability that it will end in the shortest interval of time Δt ? This probability is given by:

$$\mathbb{P}(t < T \leq t + \Delta t).$$

We can now introduce the *hazard rate* $\lambda(t)$, namely the instantaneous rate of occurrence of the event, which is defined as:

$$\lambda(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbb{P}(t < T \leq t + \Delta t | T > t)}{\Delta t}.$$

The numerator of this expression is the conditional probability that the event will occur in the interval $(t, t + \Delta t]$, given that it has not occurred before, and the denominator is the width of the interval. Dividing one by the other we obtain a rate of event occurrence per unit of time. Taking the limit as the width of the interval goes down to zero, we obtain an instantaneous rate of occurrence. Let's now rewrite the term $\mathbb{P}(t < T \leq t + \Delta t | T > t)$ as the ratio between $\mathbb{P}(t < T \leq t + \Delta t)$ (probability that T is in the interval $(t, t + \Delta t]$) and $\mathbb{P}(T > t)$. This yields:

$$\lambda(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbb{P}(t < T \leq t + \Delta t)}{\mathbb{P}(T > t) \Delta t},$$

which can finally be rewritten as:

$$\lambda(t) = \lim_{\Delta t \rightarrow 0} \frac{F(t + \Delta t) - F(t)}{S(t) \Delta t} = \frac{f(t)}{S(t)}.$$

In words, the rate of occurrence of the event at duration t equals the density of events at t , divided by the probability of surviving to that duration without experiencing the event.

3.2.2 Kaplan-Meier estimator

In 1958, Kaplan and Meier proposed a non-parametric statistic (also known as product limit estimator) to estimate the survival function from lifetime (censored) data. Denote by $t_1 \leq t_2 \leq \dots \leq t_N$ the distinct ordered times of death. Let d_i be the number of deaths at time t_i , and let n_i be the number alive (*and under consideration, i.e. not censored*) just before time t_i . This is the number exposed to risk at time t_i . Then, the Kaplan-Meier estimate of the survival function is:

$$\hat{S}(t) = \prod_{i: t_i \leq t} \left(1 - \frac{d_i}{n_i}\right).$$

A heuristic justification of the estimate is as follows. To survive until time t you must first survive until time t_1 . You must then survive from t_1 to t_2 given that you have already survived until t_1 . And so on. Since there are no deaths between t_{i-1} and t_i , we take the probability of dying between these times to be zero. The conditional probability of dying at t_i given that the subject was alive just before can be estimated by d_i/n_i . The conditional probability of surviving until time t_i is the complement $1 - d_i/n_i$. The overall unconditional probability of surviving to t is obtained by multiplying the conditional probabilities for all relevant times up to t .

Example. Consider the following remission times (in weeks) for 21 patients suffering from leukaemia: 1, 1, 2, 2, 3, 4, 4, 5, 5, 8, 8, 8, 8, 11, 11, 12, 12, 15, 17, 22, 23. We can calculate t and $\hat{S}(t)$ as follows:

t	$\hat{S}(t)$
$t < 1$	$\frac{21}{21} = 1$
$1 \leq t < 2$	$1 \cdot \frac{21-2}{21} = 0.905$
$2 \leq t < 3$	$1 \cdot 0.905 \cdot \frac{19-2}{19} = 0.81$
...	...

The result can be represented by the Kaplan-Meier curve (see Figure 1.3), a step function with discontinuities at the observed death times.

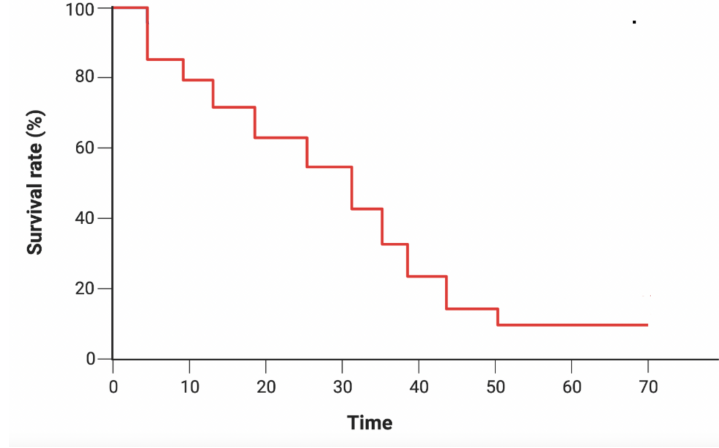


Figure 1.3: Kaplan-Meier curve: series of declining horizontal steps as time progresses.

3.2.3 Cox proportional hazards model

The Cox proportional hazards model, proposed by David Cox (1972), is a very established model in the bio-medical research domain; in fact, it was also adopted in the Framingham Heart Study. The goal of the model is to describe the hazard rate $\lambda(t)$ as a function of predictors:

$$\lambda(t|\mathbf{x}_i) = \lambda_0(t)e^{\beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}} = \lambda_0(t)e^{\boldsymbol{\beta}^\top \mathbf{x}_i},$$

where t is the survival time, $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})$ a set of predictors, $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)$ the corresponding coefficients, and λ_0 the so-called *baseline hazard*, which corresponds to the value of the hazard if all the predictors are equal to zero. A value of the i -th coefficient β_i greater than 0 indicates that as the value of the i -th predictor increases, the event hazard increases and thus the length of survival decreases. Put another way, a hazard ratio above 1 indicates a predictor that is positively associated with the event probability, and thus negatively associated with the length of survival.

Let t_i be the time at which individual i exists. The likelihood of this event is

$$\ell_i(\boldsymbol{\beta}) = \frac{e^{\boldsymbol{\beta}^\top \mathbf{x}_i}}{\sum_{j \in R(t_i)} e^{\boldsymbol{\beta}^\top \mathbf{x}_j}},$$

where $R(t_i)$ are all individuals at risk at time t_i . Assuming that there is no censoring, we can find $\boldsymbol{\beta}$ by solving the following problem:

$$\max_{\boldsymbol{\beta}} \sum_{i=1}^n \left(\boldsymbol{\beta}^\top \mathbf{x}_i - \ln \left(\sum_{j \in R(t_i)} e^{\boldsymbol{\beta}^\top \mathbf{x}_j} \right) \right),$$

where n is the number of observations. This model can be extended to censored data and instances in which multiple individuals leave at the same time (or $\mathbf{x}$ depends on t).